Implementation of Quantum Molecular Docking Using Quadratic Unconstrained Binary Optimization

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1. Introduction

The field of combinatorial Optimization (CO) represents one of the most important areas in the field of optimization, with practical applications found in every industry, including both the private and public sectors. It is one of the most active research areas pursued by the research communities of Operations Research, Computer Science and Analytics as they work to design and test new methods for solving real world CO problems. These problems typically involve making optimal decisions in setting where many yes/no must be made and each of these decisions yields a corresponding objective function value such as cost minimization or profit maximization. The computational complexity inherent in identifying optimal or near-optimal solutions in such settings presents formidable challenges, particularly as problem instances scales to realistic dimensions.

The traditional approach to solving CO problems has involved developing solution algorithms that are tailored to exploit the specific mathematical structure of the individual problem classes. While this approach has produced good results in certain problem settings, it suffers from a fundamental limitation: the diversity of applications arising in practice requires the creation of a diversity of solution techniques, each with limited application outside their original intended use. As a transformative development in recent years, we have discovered that a mathematical formulation known as QUBO, an acronym for a Quadratic Binary Optimization Problem, can embrace an exceptional variety of important CO problems found in industry. The QUBO model has emerged as an underpinning of the quantum computing area known as quantum annealing and Fujitsu's digital annealing and has become a subject of study in neuromorphic computing. QUBO models lie at the heart of experimental platforms developed by leading technology organizations including D-Wave Systems, IBM, and others, positioning them at intersections of classical and quantum computational research. A particularly valuable property of QUBO is its mathematical equivalence to the Ising model from statistical physics, effectively bridging optimization theory and physical systems, enabling bidirectional flow of ideas between these traditionally separate domains.

Among the numerous applications domains where QUBO formulations demonstrate compelling potential, computational drug discovery represents an area of particular interest. Molecular docking, a cornerstone technique in structure-based drug discovery (SBDD), involves determining potential drug candidate compounds using tertiary structures of a target protein. Protein-ligand docking is a fundamental technique in SBDD and aims to predict the binding pose and binding free energy of a small-molecule ligand interacting with a target protein. The accuracy and practical utility of docking calculations depends on two primary factors: (1) the effectiveness of the search algorithm used to explore possible binding configurations and (2) the reliability of the scoring function used to estimate the binding affinity. Over the past decade, numerous classical docking tools have been developed, including AutoDock, AutoDock Vina, and Glide, which employ heuristic optimization strategies

such as genetic algorithms, iterated local search, and multi-stage filtering to efficiently explore the vast conformational space ligand poses.

A major challenge in protein-ligand docking arises from the high dimensionality of the search space. A realistic docking calculation must account for the ligand translation and rotation, internal degrees of freedom such as bond rotations, and steric and energetic compatibility with the protein binding pocket. Exhaustive enumeration of all possible ligand configuration is computationally infeasible, and therefore most classical docking methods rely on approximate and heuristic pruning strategies, which may sacrifice optimality or completeness in complex systems.

Recent advances in quantum computing and quantum-inspired optimization have motivated new formulations of molecular docking as discrete combinatorial optimization problems. Mapping docking calculations to quadratic unconstrained binary optimization (QUBO) problems enables the use of Ising machine and quantum annealing hardware to search for low-energy solutions more efficiently than traditional heuristic approaches. QUBO formalism is attractive because it allows a single objective function whose ground state corresponds to an optimal or near-optimal docking configuration.

Within the QUBO-based docking formulations, two closely related but distinct approaches have been explored: feature atom matching (FAM) and grid-point matching (GPM). In feature Atom Matching (FAM) approaches, the ligand is represented as a collection of atoms or abstracted chemical features, and binary variables encode relationships between these ligand components and protein environments. In contrast, grid-point matching approaches discretize the three-dimensional space of the protein binding site into a finite set of spatial grid points, and docking is formulated as an assignment problem between ligand atoms or features and these grid points. This discretization transforms the continuous docking problem into a finite combinatorial optimization problem suitable for QUBO encoding. In grid-point-based formulation binary variables indicate whether a specific ligand atom or feature occupies a particular grid point. Linear terms in the QUBO represent interactions energies between ligand components and the protein at specific spatial locations, while the quadratic terms encode pairwise constraints such as steric exclusion, geometric compatibility, and molecular connectivity. Feature or atom matching formulations similarly use binary variables to represent molecular replacement and interactions but differ in how spatial and chemical information is abstracted and encoded within the optimization model.

Several studies have demonstrated the feasibility of encoding molecular docking problems using these QUBO formulations for execution on quantum or quantum-inspired hardware. These approaches establish a direct connection between molecular docking and combinatorial optimization, providing a systematic framework for exploring protein-ligand binding configurations beyond classical heuristic methods. However, challenges remain in scaling these formulations due to the large number of variables and constraints required to represent atomic-level geometry and spatial discretization accurately.

Therefore, in this study, we focus on the formulation and solution of quadratic unconstrained binary optimization problems, with particular emphasis on their application to molecular docking. First, we introduce QUBO formulation starting from general polynomial objective functions and describe the

systematic reduction of higher-order terms into quadratic form through the introduction of auxiliary variables and penalty functions. We explicitly discuss how constraints are encoded as penalty terms in the Hamiltonian to ensure feasibility of solutions with the unconstrained optimization framework. Subsequently, we extend this formulation to protein-ligand docking and present concrete molecular docking examples in which docking configurations are expressed as a QUBO problem. In this formulation, binary decision variables represent molecular placement and assignment choices, while interaction energies, geometric constraints, and steric exclusions are incorporated as linear and quadratic terms in the Hamiltonian. This real-world example illustrates how abstract QUBO construction principles translate into physically meaningful docking models.

Finally, we address the solution of the resulting QUBO problems using classical and quantum-inspired optimization tools. The formulated models are solved using mathematical optimization frameworks and solvers, including Pyomo-based implementations with SCIP and CPLEX, as well as the Hercules optimizer. The solutions obtained are analyzed to evaluate feasibility, optimality and computational performance, and the results are discussed in the context of their applicability to molecular docking and combinatorial optimization problems.

2. Basic QUBO Problem Formulation

Definition: The QUBO model is expressed by the optimization problem:

$$\text{QUBO: minimize/maximize } y = x^T Q x$$

Where x is vector of binary decision variables and Q is a square matrix of constants.

The Q matrix is assumed to be symmetric because the interaction terms satisfy $x_{ij} = x_{ji}$, making asymmetric coefficient redundant. Or in Upper triangular form to avoid double counting and for impact representation. It can be achieved as:

Symmetric form: For all I and j except $i = j$, replace q_{ij} by $(q_{ij} + q_{ji})/2$.

Upper triangular form: For all i and j with $j > i$, replace q_{ij} by $q_{ij} + q_{ji}$. Then replace all q_{ij} for $j < i$ by 0.

2.1 Illustrative Example

$$\text{Minimize } y = -5x_1 - 3x_2 - 8x_3 - 6x_4 + 4x_1x_2 + 8x_1x_3 + 2x_2x_3 + 10x_3x_4$$

Here the linear part is $-5x_1 - 3x_2 - 8x_3 - 6x_4$ and quadratic part is $4x_1x_2 + 8x_1x_3 + 2x_2x_3 + 10x_3x_4$

Since binary variables satisfy $x_j = x_j^2$, the linear part can be written as $-5x_1^2 - 3x_2^2 - 8x_3^2 - 6x_4^2$

This means linear terms can be absorbed into the quadratic form by placing them on diagonal of Q matrix.

Then we can re-write the model in the following matrix form

$$\text{Minimize } y = (x_1\ x_2\ x_3\ x_4)\begin{bmatrix} \mathbf{-5} & \mathbf{2} & \mathbf{4} & \mathbf{0} \\ \mathbf{2} & \mathbf{-3} & \mathbf{1} & \mathbf{0} \\ \mathbf{4} & \mathbf{1} & \mathbf{-8} & \mathbf{5} \\ \mathbf{0} & \mathbf{0} & \mathbf{5} & \mathbf{-6} \end{bmatrix}\begin{bmatrix} \boldsymbol{x_1} \\ \boldsymbol{x_2} \\ \boldsymbol{x_3} \\ \boldsymbol{x_4} \end{bmatrix}$$

In turn, this can be written in the matrix notation as

$$\text{Minimize } y = x^T Q x$$

Where x is column vector of binary variables. Here the coefficients of the original linear terms appear on the main diagonal of the Q matrix. In this case Q is symmetric about the main diagonal without needing to modify the coefficients.

2.2 Creating QUBO Models Using Known Penalties

The natural form of a QUBO model includes no explicit constraints other than the requirement that decision variables be binary. However, many optimization problems of practical interest include additional constraints that must be satisfied during the search for optimal solutions. Such constraints problems can be reformulated as unconstrained QUBO models by introducing quadratic penalty terms into the objective function rather than explicitly imposing constraints in the classical sense. The penalties are formulated so they equal zero for feasible solutions and equal some positive penalty amount for infeasible solutions. For a minimization problem, these penalties are added to create an augmented objective function to be minimized. If the penalty terms can be driven to zero, the augmented objective function becomes the original function to be minimized.

For certain types of constraints, quadratic penalties useful for creating QUBO models are known in advance and readily available to be used in transforming a given constrained problem into a QUBO model.

To illustrate this approach, consider a constrained optimization problem of the form

$$\text{Min } y = f(x),$$

subject to constraint

$$x_1 + x_2 \leq 1$$

Where x_1 and x_2 are binary variables. The quadratic penalty that corresponds to this constraint is

$$P x_1 x_2$$

Where P is a positive scalar penalty value. This must be chosen sufficiently large to assure the penalty term is indeed equivalent to the classical constraint. Here the unconstrained problem

$$\text{Minimize } y = f(x) + P x_1 x_2$$

has the same optimal solution as the original constrained problem. If f(x) is linear or quadratic, then this unconstrained model will be in the form of a QUBO model. In our present example, any

optimizer trying to minimize y will tend to avoid solutions having both x_1 and x_2 equal to 1, else a large positive amount will be added to the objective function.

2.3 Relationship to Ising models

QUBO is mathematically equivalent to the Ising model from statistical physics. By applying the variable transformation:

$$x_i = \frac{1 + s_i}{2},\ s_i \in \{-1, +1\}$$

the QUBO objective can be mapped to an Ising Hamiltonian:

$$H(s) = \sum_i h_i\, s_i + \sum_{i<j} J_{ij}\, s_i s_j$$

where h_i are local fields and J_{ij} are spin–spin couplings. This equivalence enables QUBO problems to be solved using quantum annealers, simulated annealing, and quantum-inspired Ising solvers.

3. Molecular Docking and QUBO

3.1 Introduction to Molecular Docking

Molecular Docking is a widely used in silico technique in virtual screening, lead optimization, and mechanism study. It determines the pose and binding free energy (ΔG_{Bind}) between a ligand and a protein. Suppose a ligand consist of n atoms (a_1, a_2, a_3……., a_n) and the coordinate of a_i is r_i, molecular docking to minimize

$$\Delta G_{bind}(a_1, a_2, \ldots., a_n\ , r_1, r_2, \ldots., r_n) r_i\ \in D \qquad (1)$$

Domain D is a docking box restricting the pose of the ligand. Equation 1 is usually called a scoring function. However, minimizing (1) is nondeterministic polynomial-time (NP-hard), suggesting that its precise solution require vast enumeration of r_i. To accelerate the molecular docking, software like AutoDockFR and GOLD apply approximation algorithms like simulated annealing. Glide applied enumeration but in a stepwise fashion, that different scoring functions, from rough to precise, are in order used to narrow the solution space step by step into the final docking poses.

Quantum computers (QC) like D-WAVE and Coherent Ising Machine (CIM) are new approaches to accelerate the solving of NP-hard problems. A 100,000 qubit CIM has been reported to solve MAX-CUT 1000 times faster than advanced digital computers. Recently they have also been explored in NP-hard problems in science like compressed sensing, RNA folding, molecular unfolding and protein protonation. To solve a NP-hard problem with QCs, one way is to encode the problem into a quadratic unconstrained binary optimization (QUBO) model which is to minimize a quadratic form ($x^T Q x$). In QCs, x is represented by the state of r qubits while the coefficient matrix (Q_{rXr}) is

translated to interactions among qubits. The qubit system has evolved to reach a ground-state Hamilton, which corresponds to minima of the QUBO instance.

3.2 Converting Molecular Docking into QUBO

To encode molecular docking into a QUBO models, we first translate molecular docking into a matching problem between ligand atoms and space positions and then describe the matching problem with a QUBO instance. Inspired by the grid methods, first we discretise D into N grid points $(g_1, g_2, \ldots, g_N)$, to translate molecular docking into a matching problem between atoms of ligands and grid points. If a_i matches grid point g_{s_i}, the distance between a_i and a_j is d_{ij} and the distance between g_{s_i} , and g_{s_i} is D, ΔG_{Bind} in eq 1 could be estimated as $\Delta G_{Bind}(a_1, a_2, \ldots, a_n, r_1, r_2, \ldots, r_n) \approx \Delta G'_{\text{Bind}}$

$$(a_1, a_2, \ldots, a_n, g_{s1}, g_{s2}, \ldots, g_{sn}) = \sum_{i=1}^{n} w_{a_i g_{s_i}}$$

$$s_i = 1, 2, \ldots, N \quad s.t. \left|d_{ij} - D_{sisj}\right| \leq C_{dist} \qquad (3)$$

where $w_{a_i g_{s_i}}$ defines the fitness when a_i is placed around g_{s_i}, and could be viewed as an intermolecular term in ΔG_{Bind}, while the constraint term retains the shape of the ligand and could be regarded as an intramolecular term in ΔG_{Bind}. Here eq 3 is converted into a QUBO model. X_{ij} is applied as the binary decision variable to indicate whether a_i and g_j is matched. Therefore, eq 3 could be interpreted as a quadratic form in (4). Note that an additional constraint term is included to restrict that one atom should match at most one grid point. However, one grid point could match to more than one atom because of the discretization of the docking box D.

$$\Delta G'_{\text{Bind}}\,(a_1, a_2, \ldots \ldots, a_n, g_{s1}, g_{s2}, \ldots, g_{sn}$$

$$= \Delta G'_{\text{Bind}}\,(a_1, a_2, \ldots, a_n, g_1, g_2, \ldots, g_N, x_{11}, x_{12}, \ldots, xnN)$$

$$= \sum_{i=1}^{n} \sum_{j=1}^{N} w_{i,g_j}\, x_{ij}^2,$$

$$x_{ij} = \begin{cases} 1, & a_i \text{ matches } g_j \\ 0, & a_i \text{ does not match } g_j \end{cases}$$

$$s.t\, \left|d_{ik} - D_{jl}\right| \leq c_{dist}\ and \sum_{i=1}^{n} x_{ij} \leq 1 \qquad (4)$$

Unlike the QUBO in eq 2, constraints exist in the current QUBO model. Therefore, we then converted constraints into two quadratic Lagrange terms in eq 5. K_{dist}, K_{mono}, and C_{dist} are three parameters that require parametrization.

$$\Delta G''_{\text{Bind}}\,(a_1, a_2, \ldots, a_n, g_1, g_2, \ldots \ldots, g_N, x_{11}, x_{12}, \ldots \ldots, x_{nN})$$

$$= \sum_{i=1}^{n} \sum_{j=1}^{N} w_{a_i g_j} x_{ij}^2 + K_{dist} \sum_{i=1}^{n} \sum_{j=1}^{N} \sum_{k=1}^{n} \sum_{l=j+1}^{N} u_{ijkl}\, x_{ij} x_{kl}$$

$$+ K_{mono} \sum_{i=1}^{n} \sum_{j=1}^{N} \sum_{k=1}^{n} \sum_{l=j+1}^{N} v_{ijkl}\, x_{ij} x_{kl}$$

$$= \begin{pmatrix} x_{11} \\ x_{12} \\ \cdots \\ x_{nN} \end{pmatrix}^T \begin{pmatrix} w_{a_1 g_1} & K_{dist}\, u_{1112} + K_{mono}\, v_{1112} & \cdots & K_{dist}\, u_{11nN} + K_{mono}\, v_{11nN} \\ 0 & w_{a_1 g_2} & \cdots & K_{dist}\, u_{12nN} + K_{mono}\, v_{12nN} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & w_{a_n g_N} \end{pmatrix} \begin{pmatrix} x_{11} \\ x_{12} \\ \cdots \\ x_{nN} \end{pmatrix}$$

$$u_{ijkl} = \begin{cases} 1, & |\, d_{ik} - D_{jl} \,| > c_{dist} \\ 0, & |\, d_{ik} - D_{jl} \,| > c_{dist} \end{cases} \qquad v_{ijkl} = \begin{cases} 1, & j = l \text{ and } i \neq k \\ 0, & \text{otherwise} \end{cases}$$

$$x_{ij} = \begin{cases} 1, & a_i \text{ matches } g_j \\ 0, & a_i \text{ does not match } g_j \end{cases} \qquad (5)$$

The QUBO models could then be solved by Quantum Computers, as mentioned above. Due to their stochastic nature, QCs could achieve different results with different runs. These results are taken as sampled poses of ligand binding. Those poses could be further ranked via comparison of estimated ΔG_{Bind}. Finally, solved matches are converted to docking poses. As proof of concept, only the rigid docking considered where the shape of the ligand is maintained. Therefore, the conversion is realized via an alignment. Suppose the standardized coordinate of a_i is r_i^{st} and it matches to grid point g_{s_i}, whose coordinate is R_{s_i}, a Kabsch RMSD rotation matrix is calculated to be

$$K_{\text{rot}} = \arg\min_{K} \left[K \begin{pmatrix} r_1^{st} \\ r_2^{st} \\ \cdots \\ r_n^{st} \end{pmatrix} + b - \begin{pmatrix} R_{g_1} \\ R_{g_2} \\ \cdots \\ R_{g_n} \end{pmatrix} \right], \text{ where}$$

$$b = \frac{\sum_{i=1}^{n} \mathrm{R}_{g_i}}{n} - \frac{\sum_{i=1}^{n} \mathrm{R}_i^{\mathrm{st}}}{n} \qquad (6)$$

b is a column vector unifying the centre of ligand atoms and their matched grid points. The final docking pose is calculated by eq 7.

$$\begin{pmatrix} \mathrm{r}_1^{\mathrm{dock}} \\ \mathrm{r}_2^{\mathrm{dock}} \\ \dots \\ \mathrm{r}_n^{\mathrm{dock}} \end{pmatrix} = K_{\mathrm{rot}} \begin{pmatrix} \mathrm{r}_1^{\mathrm{st}} \\ \mathrm{r}_2^{\mathrm{st}} \\ \dots \\ \mathrm{r}_n^{\mathrm{st}} \end{pmatrix} + \mathrm{b} \qquad (7)$$

To quantify $w_{a_i g_i}$ in eq 5 , two encoding methods can be used to generate grid points, Grid Point Matching (GPM) and Feature Atom Matching (FAM).

3.3 Grid Point Matching (GPM)

In Grid Point Matching (GPM), grid points are directly generated in a predefined docking box D with a gap of 2 Å. $w_{a_i g_i}$ is defined as the van der Waals energy when the atomic type of a_i is placed on g_i and is precalculated and saved on each g_i using AutoGrid in AutoDockFR. In fact, in the scoring function of AutoDockDFR, electrostatic and solvation terms are also included. They are neglected due to poorer performances possibly accounted for by accumulation of error in discrete space. Matches whose $w_{a_i g_i}$ is above 0 are dropped for reducing the consumption of qubits.

3.4 Feature Atom Matching (FAM)

In Feature Atom Matching (FAM), grid points generated with a gap of 1 Å in the predefined docking box coarse-grained into 3 types of "feature atoms" (Hydrophobic(C), Hydrogen bond donor (HD), hydrogen bond acceptor (OA)) using AutoSite algorithm. $w_{a_i g_i}$ in FAM is defined as $|X_{a_i} - X_{g_i}| - 0.5$, where X_{a_i} and X_{g_i} is the Pauling's electronegativity of the ligand atom and FAs, respectively.

Here, Molecular docking is split into pose sampling and pose scoring. Pose sampling is to sample possible solutions of the docking pose while pose scoring is to rank these poses. Comparatively, pose sampling is NP-hard while pose scoring is not, it only requires a rough scoring function. Therefore, GPM and FAM are used for pose sampling while pose scoring could be left for classical computers.

Note: GPM and FAM are suitable for docking small molecules and peptides but not protein-protein docking.

3.5 Limitation of GPM and FAM encoding

The GPM and FAM encoding are limited by the quadratic growth of qubit with respect to the number of ligand atoms. Benchmarking shows that maximum feasible input size are approximately 156 atoms for GPM and 305 atoms for FAM, which corresponds to peptides of about 15 and 30 residues respectively. Protein-Protein docking involves more atoms resulting in requirement of greater number of qubits exceeding the capacity of current coherent Ising machines. In addition, both encoding is based on discretization of the docking box, and the sampling error exhibits a strong linear dependence on discretization error. Reducing this error require spatial discretization, which further increases the number of binary variables and qubits. Therefore, the encoding cannot be scaled to large docking systems without exceeding available quantum computational resources.

4. Implementation Details and Result Analysis

The QUBO formulations in this work were taken as concrete optimization problems and solved using a combination of exact mixed-integer optimization solvers-SCIP and CPLEX and heuristic QUBO

solvers. The computational study covers both small benchmarks QUBO instances and large, structured QUBOs arising from molecular docking formulations.

Exact solvers were employed to establish reference solutions and to analyse optimality a solution degeneracy, while heuristic solver was used to examine scalability and convergence behaviour under limited search. In addition to objective values, the resulting binary configurations were analysed to assess structural stability and overlap among solutions obtained by different solvers. This computational framework allows systematic comparison of solver behaviour as problem size, coupling structure, and constraint density increase.

4.1 Solvers

Exact optimization was performed using SCIP and CPLEX, accessed through a Pyomo-based modelling framework. The QUBO formulations were instantiated in Pyomo as binary quadratic optimization models and passed to the solvers in mixed-integer programming (MIP) form.

Pyomo: A python-based open-source software package that supports a diverse set of optimization capabilities for formulating and analysing optimizations models. Pyomo can be used to define symbolic problem, create concrete problem instances, and solve these instances with standard solvers. It supports analysis and scripting within a full-featured programming language. Furthermore, Pyomo has also proven an effective framework for developing high-level optimization and analysis tools.

SCIP: It is one of the fastest academically developed solvers for mixed integer programming (MIP) and mixed integer nonlinear programming (MINLP). In addition, SCIP provides a highly flexible framework for constraint integer programming, branch-cut-and-price, and can optionally be configured to solve mixed-integer linear programs in a numerically exact solving mode. SCIP allows for total control of the solution process and access of detailed information down to the guts of the solver.

CPLEX: It is a state-of-the-art commercial solver for large scale mixed-integer programming (MIP) and quadratic optimizations problems, widely accepted in both academic research and industrial applications. It provides highly optimized implementations of branch-and-bound, branch-and-cut, and cutting-plane algorithms, along with advanced presolve techniques. CPLEX supports mixed-integer linear programming (MILP), mixed-integer quadratic programming (MIQP), and quadratically constrained programming, with robust numerical stability and precision. It allows for total control of the solution process and access of detailed information down to the guts of the solver.

Hercules

Hercules is a Rust library for analyzing, finding approximate solutions, and finding exact solutions for Quadratic Unconstrained Binary Optimization (QUBO) problems. Hercules is designed as a simple, easy-to-use library for exactly or approximately solving QUBO problems. It is not necessarily designed to be a state-of-the-art tool but a toolkit for quickly prototyping and testing QUBO methods. That said, Hercules is designed to be fast and written in Rust, a high-performance systems programming language. It is still experimental and currently in the early stage of development.

Qubolite

QUBOLite is a lightweight Python framework for formulating and solving Quadratic Unconstrained Binary Optimization (QUBO) problems. It supports exact brute-force optimization for small and moderate problem sizes, making it suitable as a ground-truth reference for validating heuristic and solver-based approaches. Its simple NumPy-based design enables direct evaluation of QUBO energies and clear analysis of optimal bitstring structures.

4.2 Degeneracy and Multiple Global Optima in QUBO Models

In QUBO, a degenerate state refers to any specific binary configurations (x) that resides at the lowest possible energy level of the objective function. When a problem possesses more than one such state, it is said to have Multiple Global Optima.

Mathematically, a QUBO problem is defined as the minimization of a quadratic form

$$\min_{x \in \{0,1\}^n} x^\top Q x.$$

If two distinct binary vectors $x^{(1)} \neq x^{(2)}$ satisfy

$$\left(x^{(1)}\right)^\top Q x^{(1)} = \left(x^{(2)}\right)^\top Q x^{(2)},$$

then both configurations constitute globally optimal solutions. The QUBO formulation guarantees minimization of the objective value but does not impose uniqueness of the minimizing configuration.

Several structural properties of QUBO models contribute to this phenomenon:

1. **Symmetry in the QUBO formulation**
   In many problems, certain variables or groups are interchangeable. Swapping these variables does not alter the quadratic energy, resulting in distinct bitstrings with identical objective values.

2. **Flat energy landscapes**
   Large, highly coupled QUBO matrices often exhibit flat regions (plateaus) near the global minimum. This is common in dense QUBOs where competing interactions balance each other out exactly allowing multiple configurations to share the same minimum energy.

3. **Degeneracy induced by constraint penalties**

   Penalty-based constraint encoding can introduce degeneracy when multiple variable combinations satisfy constraints equally well while contributing the same net energy to the system.

4.3 Experimental design and Benchmark Selection

Two categories of QUBO instances were considered:

1. **Benchmark QUBO Matrices**, consisting of small to moderate-sized instances (4×4, 5×5, and 8×8), used to validate solver correctness and analyse baseline behaviour.

2. **Application-Specific QUBO**, representing a molecular docking QUBO formulation derived via GPM and FAM encoding method.

For each instance, the final objective value, optimal binary decision vector, and structural overlap between solver solutions were recorded.

4.4 Results on Benchmark QUBO Matrices.

a. Objective Value and Optimality Analysis

Table 1: Matrices

S.N.	Matrices	Size
1.	$Q_1 = \begin{bmatrix} -5 & 4 & 8 & 0 \\ 4 & -3 & 2 & 0 \\ 8 & 2 & -8 & 10 \\ 0 & 0 & 10 & -6 \end{bmatrix}$	4 x 4
2.	$Q_2 = \begin{bmatrix} -2 & 1 & 1 & 0 & 0 \\ 1 & -2 & 0 & 1 & 0 \\ 1 & 0 & -3 & 1 & 1 \\ 0 & 1 & 1 & -3 & 1 \\ 0 & 0 & 1 & 1 & -2 \end{bmatrix}$	5 x 5
3.	$Q_3 = \begin{bmatrix} -2 & 0 & 0 & 0 & 3 & -2 & 2 & -3 \\ 0 & 1 & 0 & 0 & -2 & -1 & -2 & 3 \\ 0 & 0 & 2 & 0 & -3 & 2 & 2 & 3 \\ 0 & 0 & 0 & -1 & 1 & 1 & -1 & -1 \\ 3 & -2 & -3 & 1 & -1 & 0 & 0 & 0 \\ -2 & -1 & 2 & 1 & 0 & 3 & 0 & 0 \\ 2 & -2 & 2 & -1 & 0 & 0 & -3 & 0 \\ -2 & 3 & 3 & -1 & 0 & 0 & 0 & 1 \end{bmatrix}$	8 x 8
4.	$Q_4 = \begin{bmatrix} -3525 & 175 & 325 & 775 & 1050 & 425 & 525 & 250 \\ 175 & -1113 & 91 & 217 & 294 & 119 & 147 & 70 \\ 325 & 91 & -1989 & 403 & 546 & 221 & 273 & 130 \\ 775 & 217 & 403 & -4185 & 1302 & 527 & 651 & 310 \\ 1050 & 294 & 546 & 1302 & -5208 & 714 & 882 & 420 \\ 425 & 119 & 221 & 527 & 714 & -2533 & 357 & 170 \\ 525 & 147 & 273 & 651 & 882 & 357 & -3045 & 210 \\ 250 & 70 & 130 & 310 & 420 & 170 & 210 & -1560 \end{bmatrix}$	8 x 8

Table 2: Objective Value and Optimal Analysis

S.N.	Matrices	SCIP		CPLEX		Hercules		Qubolite	
		Active Bitstring	Optimal obj	Active Bitstring	Optimal obj	Active Bitstring	Optimal obj	Active Bitstring	Optimal obj

			Value		Value		Value		Value
1.	Q_1	[1 0 0 1]	-11.0	[1 0 0 1]	-11.0	[1 0 0 1]	-0.00059	[1 0 0 1]	-11.0
		[0 1 0 1]	-9.0	[0 1 0 1]	-9.0	-	-	-	-
		[0 0 1 0]	-8.0	[0 0 1 0]	-8.0	-	-	-	-
		[0 1 1 0]	-7.0	[0 1 1 0]	-7.0	-	-	-	-
2.	Q_2	[0 1 1 0 0]	-5.0	[1 0 0 1 0]	-5.0	[1 1 1 0 1]	-0.0003	[1 0 0 1 0]	-5.0
		[1 0 0 1 0]	-5.0	[1 1 1 0 1]	-3.0	-	-	-	-
		[1 1 1 0 1]	-3.0	[0 1 1 0 0]	-5.0	-	-	-	-
		[1 0 0 1 0]	-5.0	-	-	-	-	-	-
3.	Q_3	[0 1 0 1 1 0 1 0]	-12.0	[0 1 0 1 1 0 1 0]	-12.0	[0 1 0 1 10 1 0]	-0.00039	[0 1 0 1 1 0 1 0]	-12.0
		[0 1 1 1 1 0 1 0]	-12.0	[0 1 1 1 1 0 1 0]	-12.0	-	-	-	-
		[0 1 0 0 1 0 1 0]	-11.0	[0 1 0 0 1 0 1 0]	-11.0	-	-	-	-
		[1 0 0 1 0 0 1 1]	-11.0	[0 1 1 0 1 0 1 0]	-11.0	-	-	-	-
4.	Q_4	[0 0 0 1 1 0 0 1]	-6889.0	[1 1 1 0 0 11 0]	-6889.0	[1 1 0 1 1 1 1 1]	-0.5789	[1 0 0 1 0 1 0 1]	-6889.0
		[1 0 0 1 0 1 0 1]	-6889.0	[1 0 0 1 0 1 0 1]	-6889.0	[0 0 0 1 1 0 0 1]	-0.8921	-	-
		[1 1 0 1 1 1 1 1]	-1989.0	[1 1 0 1 1 1 1 1]	-1989.0	[1 1 1 0 0 1 1 0]	-0.9547	-	-
		[1 1 1 0 0 1 1 0]	-1989.0	[0 0 0 1 1 0 0 1]	-6889.0	-	-	-	-

White = SCIP, Green = Hercules, Red = CPLEX, Purple=Qubolite

Across all tested QUBO instances $(Q_1 - Q_4)$, the solvers SCIP, CPLEX and Qubolite consistently converged to identical optimal objective values. For instances Q_1, Q_3 *and* Q_4 , these solvers recovered multiple structurally distinct bitstrings that yield the same minimum energy, empirically confirming the **degeneracy.** Specifically, in Q_3 and Q_4, the detection of different bitstrings (e.g. [0,1,0,1,1,0,1,0] vs. [0,1,1,1,1,0,1,0] in Q_3) with a shared objective value of -12.0 demonstrates the presence of multiple global optima within the energy landscape.

b. Structural Bitstring Analysis

Objective value agreement does not uniquely characterize solver behavior. To further investigate the solution space, the structural overlap between binary solutions was analyzed. This analysis is crucial for identifying whether solvers are converging to the same specific bitstring or finding distinct **degenerate states** (Multiple Global Optima).

Table 3: Structural bitstring analysis

S. No.	Matrix Size	Solver Pair	Common Active Bits (Position)	Overlap Count	Structural Agreement (%)
1	Q_1 $(4\ x\ 4)$	SCIP – CPLEX	{1, 4}	2 / 2	100
		SCIP -Qubolite	{1, 4}	2 / 2	100
		SCIP – Hercules	{1, 4}	2 / 2	100
		CPLEX – Qubolite	{1, 4}	2 / 2	100
		CPLEX – Hercules	{1, 4}	2 / 2	100
		Hercules - Qubolite	{1, 4}	2 / 2	100
2	Q_2 $(5\ x\ 5\ )$	SCIP – CPLEX	∅	0 / 2	0
		SCIP - Qubolite	∅	0 / 2	0
		SCIP – Hercules	{2, 3}	2 / 2	100
		CPLEX - Qubolite	{1,4}	2/2	100
		CPLEX – Hercules	{1}	1 / 2	50
		Hercules- Qubolite	{1}	1/2	50
3	Q_3 $(8\ x\ 8)$	SCIP – CPLEX	{2,4,5,7}	4 / 4	100
		SCIP – Qubolite	{2,4,5,7}	4 / 4	100
		SCIP – Hercules	{2,4,5,7}	4 / 4	100
		CPLEX – Qubolite	{2,4,5,7}	4 / 4	100
		CPLEX – Hercules	{2, 4, 5, 7}	4 / 4	100
		Hercules - Qubolite	{2,4,5,7}	4 / 4	100
4	Q_4 $(8\ x\ 8)$	SCIP – CPLEX	∅	0 / 3	0
		SCIP – Qubolite	{4, 5, 8}	3 / 3	100
		SCIP – Hercules	{4, 5, 8}	3 / 3	100
		CPLEX – Qubolite	∅	0 / 3	0
		CPLEX – Hercules	{1, 2, 6, 7}	4 / 5	80
		Hercules - Qubolite	{4, 5, 8}	3 / 3	100

Analysis of Structural Convergence

The comparison of active bit positions across the four instances reveals distinct behaviors regarding solution uniqueness and degeneracy:

- **Global Uniqueness Q_1 and Q_3 :** For the $4\ x\ 4$ (Q_1) and the first $8\ x\ 8$ (Q_3) instances, all solvers exhibit Perfect Structural Agreement. The identical active bit positions across SCIP, CPLEX, QUBOLite, and Hercules suggest that these problems possess a unique global optimum, or that the solvers' heuristics are biased toward the same configuration.
- **Empirical Degeneracy (Q_2) :** In the $5\ x\ 5$ instance, a clear divergence in structural agreement is observed. While SCIP, CPLEX, and QUBOLite all reached the optimal objective value of -5.0, the Zero Overlap between SCIP and CPLEX/QUBOLite confirms the existence of Multiple Global Optima. This instance captures two structurally distinct degenerate states that satisfy the optimization criteria equally.
- **Structural Divergence and Local Minima (Q_4) :** In the $8\ x\ 8$ (Case-2) instance, the structural analysis highlights complex behavior. The lack of common active bits between SCIP and CPLEX, despite matching optimal objective values, further validates the presence

of high degeneracy in larger instances. Hercules shows a structural alignment with SCIP's pattern but fails to reach the global minimum energy, indicating that it is trapped in a local minimum that shares a similar structural "backbone" with a global ground state.

The results demonstrate that as problem complexity increases from Q_1 to Q_4, the likelihood of encountering structurally distinct but energetically equivalent solutions (degenerate states) increases, requiring bitstring analysis to fully differentiate solver performance.

c. Exact Solver Consistency and Solution Degeneracy

To quantify the structural relationship between solutions found by the exact solvers (SCIP and CPLEX), the **Jaccard Similarity Index** was calculated for each instance. This metric assesses the ratio of the intersection to the union of active bit positions, providing a mathematical measure of solution uniqueness.

Table 4: Exact Solver Consistency and Solution Degeneracy

Matrix Size	**SCIP Active Bits**	**CPLEX Active Bits**	**Common Bits**	**Jaccard Similarity**
Q_1 $(4\ x\ 4)$	{1, 4}	{1, 4}	2	1.00
Q_2 $(5\ x\ 5)$	{2, 3}	{1, 4}	0	0.00
Q_3 $(8\ x\ 8)$	{2, 4, 7, 8}	{2, 4, 5, 7}	4	1.00
Q_4 $(8\ x 8)$	{4, 5, 8}	{1, 2, 3, 6, 7}	0	0.00

Analysis of Degeneracy via Similarity

The Jaccard Similarity results provide empirical evidence of the solution landscapes discussed in Section 4.2:

- **Unitary Global Optima (J=1.00):** In the $4\ x\ 4$ and $8\ x\ 8$ (Case-1) instances, both solvers converged to the identical bitstring. This perfect similarity suggests that these specific QUBO formulations likely possess a unique global minimum or a highly dominant configuration.
- **Verification of Degeneracy (J=0.00):** For the $5\ x\ 5$ and $8\ x\ 8$ (Case-2) instances, the Jaccard Similarity is zero, despite both solvers reaching the exact same optimal objective value. This confirms that the problem is highly degenerate, admitting multiple global optima. The solvers traversed different computational paths to settle in structurally independent but energetically equivalent basins.

The results confirm that while exact solvers guarantee the optimality of the objective value, they do not enforce the uniqueness of the configuration. In industrial contexts, this diversity is advantageous, as it offers alternative operational setups that all achieve the same peak performance.

4.5 Molecular Docking QUBO: Large-Scale Application

a. Problem Characteristics

The molecular docking-QUBO represents a large, sparse optimization problem constructed for the protein structure 1Y6R using the Feature Atom Matching (FAM) and Grid Point Matching (GPM) encoding method described in the preceding sections. The formulation yields a high-dimensional binary quadratic model comprising **675** variables, producing a strongly constrained optimization landscape with significant combinatorial complexity. Such structural characteristics impose substantial computational challenges for exact solution methods, particularly with respect to search space exploration and convergence time.

b. Solver Performance and Objective Values

Table 5: Result of QUBO solved using SCIP, CPLEX, hercules

Method	**Best Obj Value**	**Number of Active Vars- 1's**	**Runtime**	**Active bitstring (1-based indices)**	**Rank**
SCIP	-8.57	18	>2 hours	[22, 35, 63, 85, 120, 233, 258, 304, 337, 368, 394, 419, 457, 474, 522, 524, 620, 650]	1
CPLIX	-8.57	18	>2 hours	[18, 35, 63, 120, 233, 259, 304, 337, 368, 394, 419, 457, 474, 522, 524, 589, 620, 650]	2
Hercules: Branch and bound	-3.2350	17	10.0319	[65, 87, 122, 137, 236, 261, 276, 303, 374, 429, 469, 489, 514, 526, 566, 592, 622]	3

The molecular docking instance provides a clear demonstration of computational complexity and structural degeneracy in large-scale QUBO models:

- **Evidence of Degeneracy:** Both SCIP and CPLEX converged to the exact same optimal objective value of **-8.57**. However, the recovered bitstrings are distinct (e.g., bit 22, 85, 258 in SCIP vs. 18, 259, 589 in CPLEX). This confirms that the docking energy landscape contains **Multiple Global Optima**, representing different molecular poses that yield the same binding energy.
- **Computational Intensity:** The runtime for exact solvers exceeded two hours, illustrating the exponential growth in search space for docking problems. While both solvers reached the global minimum, the structural variation in their results underscores that exact solvers do not

guarantee a unique solution configuration in highly symmetric or constrained chemical spaces.

- **Heuristic Efficiency vs. Accuracy:** Hercules produced a solution in approximately 10 seconds, offering a significant runtime advantage. However, the objective value (**-3.235**) is substantially higher than the global minimum. This indicates that the current Branch and Bound implementation in Hercules settled in a distant **local minimum**, highlighting a trade-off between rapid convergence and the ability to navigate the complex constraints of molecular docking QUBOs.

c. Structural Overlap Analysis

Table 6: structural overlap analysis

S. No.	Solver Pair	Common Active Bits	Overlap Count	% Overlap	Unique to Solver A	Unique to Solver B
1	SCIP vs CPLEX	{35, 63, 120, 233, 304, 337, 368, 394, 419, 457, 474, 522, 524, 620, 650}	**15**	**83.3%**	{22, 85, 258}	{18, 259, 589}
2	SCIP vs Hercules	0 (No overlap)	**0**	**0%**	All 18 bits	All 17 bits
3	CPLEX vs Hercules (B&B)	0 (No overlap)	**0**	**0%**	All 18 bits	All 17 bits

The structural data provides a critical view of the solution space for large-scale docking QUBOs:

- **Stable Degeneracy (SCIP vs CPLEX):** The high overlap (83.3%) between SCIP and CPLEX suggests a strong solution stability for the global minimum. Both solvers identified the same core structural "backbone" of 15 active bits. However, the presence of 3 unique bits for each solver—despite identical objective values (-8.57)—empirically confirms that the docking landscape contains degenerate global optima where small variations in molecular orientation do not affect the total binding energy.
- **Topological Divergence (Hercules):** The 0% overlap between Hercules and the exact solvers indicates that Hercules converged to an entirely different region of the energy landscape. This suggests that the local minimum found by Hercules (-3.235) is not structurally related to the global optima, representing a completely distinct, sub-optimal molecular pose.
- **Implications for Docking:** These results demonstrate that while exact solvers reliably find the "global neighbourhood" of the optimal docking pose, the specific bitstrings recovered can vary due to degeneracy. In contrast, the heuristic performance of Hercules highlights the difficulty of navigating towards the global basin in high-dimensional, highly constrained QUBO spaces.

4.6 Summary of Experimental Findings

The experimental evaluation conducted in this work leads to the following key conclusions regarding solver performance, structural stability, and the nature of the QUBO energy landscape:

- **Reliability of Exact Solvers:** Exact solvers, specifically SCIP and CPLEX, consistently identified the global minimum across all benchmark and application-level QUBO instances. The identical objective values achieved (e.g. (-8.57) for molecular docking) validate the correctness of the QUBO formulations and confirm that exact optimization remains the gold standard for verifying ground-state energies in small-to-moderate scales.
- **Empirical Validation of Degeneracy:** High objective-value agreement was frequently accompanied by structural divergence in bitstrings, as evidenced by the 0.00 Jaccard Similarity in instances Q_2 and Q_4. This confirms that industrial QUBO models—particularly those with symmetric or redundant constraints—possess Multiple Global Optima. This degeneracy is a physical property of the problem landscape rather than a solver inconsistency.
- **Structural Stability in Molecular Docking:** In the high-dimensional molecular docking QUBO, SCIP and CPLEX exhibited an 83.3% structural overlap. This indicates a high degree of solution stability, where the solvers identify a consistent "structural backbone" (15 common bits) while differing only in minor degenerate configurations. This suggests that the global basin is well-defined but contains multiple equivalent poses.
- **Heuristic Limitations and Topological Divergence:** While Hercules demonstrated superior computational speed (e.g., 10 seconds vs. >2 hours for docking), it frequently terminated at local minima. The 0% structural overlap between Hercules and the exact solver in complex instances indicates a complete "Topological Divergence," where the heuristic fails to enter the global basin and settles in an unrelated region of the solution space.
- **Multi-Dimensional Evaluation Metrics:** The study proves that objective value alone is an insufficient metric for solver evaluation. Incorporating bit-level structural analysis and overlap metrics is essential for assessing the physical relevance and reproducibility of solutions, especially in sensitive applications like molecular docking where bit positions represent specific spatial configurations.
- **Scaling Challenges in Docking QUBOs:** QUBO formulations derived from molecular docking present unique challenges, including high dimensionality and long convergence times for exact methods. The trade-off observed between the precision of SCIP/CPLEX, and the speed of Hercules emphasizes the need for hybrid optimization strategies to navigate dense, highly constrained interaction matrices.

5. Conclusion

This work presented a systematic formulation and solver-based analysis of Quadratic Unconstrained Binary Optimization (QUBO) models, with a particular emphasis on their application to molecular docking. By encoding higher-order constraints through quadratic penalty terms, we developed a framework capable of representing complex spatial and geometric constraints within a unified quadratic objective.

The computational evaluation demonstrated that exact mixed-integer solvers, such as SCIP and CPLEX, consistently identify globally optimal objective values across all tested instances. For the molecular docking formulation, both solvers reached a shared global minimum of -8.57, confirming

the mathematical integrity of the QUBO encoding. However, structural analysis revealed that identical objective values do not imply unique configurations. The observation of distinct bitstrings for the same energy level—quantified by Jaccard Similarity scores as low as 0.00 in benchmark cases—empirically validates the presence of inherent degeneracy within the QUBO energy landscape.

In the molecular docking case study, the exact solvers exhibited a high structural overlap (83.3%), suggesting a stable "structural backbone" for the global optimum, while still allowing for degenerate variations in specific bit positions. Conversely, the heuristic-based Hercules solver showcased a significant trade-off between speed and precision. While Hercules provided near-instantaneous results, the 0% structural overlap and higher objective values relative to exact solvers indicate a topological divergence, where the heuristic settles in sub-optimal local minima structurally unrelated to the global ground state.

The challenges encountered in the docking instances—specifically high dimensionality and long convergence times exceeding two hours for exact solvers—underscore the scalability limits of classical exact optimization. These findings prove that objective value alone is insufficient for evaluating QUBO solutions in application-driven contexts. A comprehensive evaluation must incorporate structural stability, solution overlap, and degeneracy analysis to ensure the physical relevance of the results, such as protein–ligand binding poses.

Ultimately, this study provides a robust benchmark for QUBO-based molecular docking and highlights the necessity of navigating degenerate solution spaces. The results motivate future research into hybrid classical–quantum optimization, quantum-inspired heuristics, and advanced constraint-reduction strategies to enhance the scalability and practical applicability of QUBO in computational biology and beyond.

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